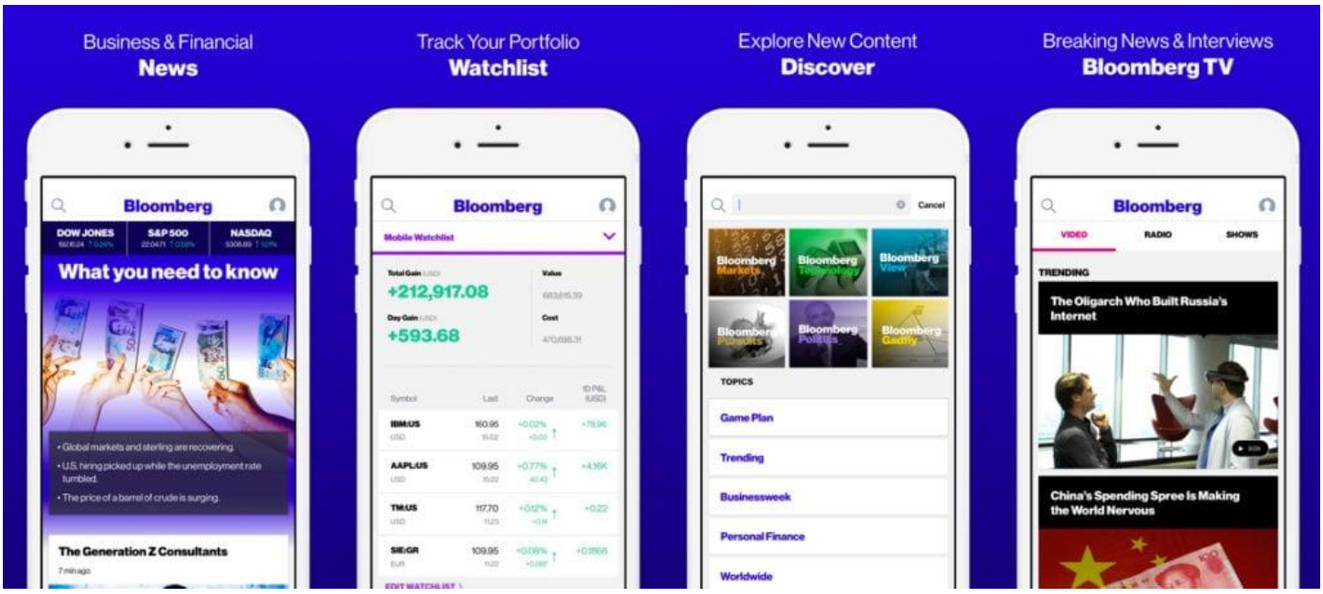

What is React Native?

React Native is a **JavaScript framework** used to build **mobile applications** that run on **Android and iOS** using **one shared codebase**.

- Created by **Meta** (Facebook)
- Uses **JavaScript**
- Based on **React**

Instead of building **two separate apps** (Java/Kotlin for Android and Swift for iOS), React Native lets you write **one app**.

Traditional Mobile Development

Platform	Language
Android	Java / Kotlin
iOS	Swift / Objective-C

❌ **Problems:**

- Two teams
- Two codebases
- More time & cost

React Native Solution

✓ One language
✓ One codebase
✓ Faster development
✓ Lower cost

Feature	React	React Native
Used for	Web apps	Mobile apps
Output	HTML	Native UI
Components	div, span	View, Text
Styling	CSS	StyleSheet

Common Components

- View → container (like div)
- Text → display text
- Image
- ScrollView
- TextInput
- Button

Navigation in React Native

Mobile apps need **screens**, not pages.

Popular library:

- **React Navigation**

Types:

- Stack Navigation
- Tab Navigation
- Drawer Navigation

For Deployment :

We use **Expo** :

✓ No Android Studio needed

✓ Fast setup

✓ Uses Expo Go app

Environment Setup (Expo Way)

Install:

- **Node.js**
- npm (included)
- Expo CLI (via Google Play Store)

-Scan QR using Expo Go App

Accessing Device Features: React Native can use:

- Camera
- GPS
- Storage
- Sensors

Backend & Database Options: Common combinations:

- Firebase
- REST API (Node.js, .NET, PHP)
- SQLite (offline apps)

React Native is **frontend** only.

Testing & Debugging

Tools:

- Expo DevTools
- Console logs
- Chrome DevTools

Installation:

1. Install Node.js : **i.** Install LTS version	Download from: ***nodejs.org***
2. Check installation version	`node -v` `npm -v`
3. Install Expo CLI	`npm install -g expo-cli`

4. Install Expo Go App on Your Phone Android → Play Store iOS → App Store ***This lets your phone act as the emulator.***	Expo Go Expo Project

5. Create Your First React Native Project	```expo init MyFirstApp```
	``cd MyFirstApp
	npm start```

6. A QR Code will appear **Run the App on Your Phone** **Open Expo Go** - **Scan the QR code** - **App loads on your phone** **✓ Live reload enabled** **✓ Changes update instantly**	